

DSA Day 38: Insertion & Shell Sort

Book: Data Structures Through C in Depth - Ch8 §8.10-8.11 · pp 427-434 | Code in Python

Overview

Insertion sort builds a sorted prefix by shifting; Shell sort generalises it with decreasing gaps.

Key concepts to master

- shifting
- gap sequence
- near-sorted efficiency

Python implementation

```
def insertion(a):  
    for i in range(1, len(a)):  
        k=a[i]; j=i-1  
        while j>=0 and a[j]>k: a[j+1]=a[j]; j-=1  
        a[j+1]=k
```

Complexity

Insertion $O(n^2)$ worst, $O(n)$ near-sorted; Shell $\sim O(n^{1.3})$.

Common pitfalls

- Wrong gap loop in Shell
- Off-by-one in the shift

Practice

- Sort with gaps 4,2,1
- Why insertion sort suits near-sorted data

